

Invoice & Payment Follow-up Automation

n8n Cloud Workflow — Technical Walkthrough

Abdul Rehman Ali | Air University Kharian | abdulrehman.tp.786@gmail.com
[linkedin.com/in/abdul-rehman-ali](https://www.linkedin.com/in/abdul-rehman-ali) | abdulrehmanali.netlify.app

Project: A fully automated invoice payment follow-up system built with n8n Cloud. The system reads unpaid invoices from Google Sheets daily, checks due dates, routes invoices into three categories (upcoming, due today, overdue), sends personalized reminder emails via Gmail, and logs all activity to a Reminders Log sheet automatically.

1 Workflow Architecture

The pipeline runs on a daily schedule. Each node handles one specific responsibility in the payment follow-up process.

Step	Node	Purpose
1	Daily Schedule	Fires every 24 hours automatically
2	Read Invoices	Reads all rows from the Invoices tab in Google Sheets
3	Check Due Dates	Code node — calculates days difference and assigns reminder type for each unpaid invoice
4	Check Skip	Filter node — stops pipeline if no invoices need reminders today
5	Route by Reminder Type	Switch node — routes invoices to correct email branch based on reminder_type
6	Send Upcoming Reminder	Gmail — sends friendly 3-5 day advance payment reminder
7	Send Due Today Reminder	Gmail — sends urgent same-day payment notice
8	Send Overdue Reminder	Gmail — sends overdue notice with days past due
9	Log to Reminders Sheet	Connected directly from Switch node — logs all invoice data with sent timestamp

2 Architecture Decisions & Rationale

Daily Schedule Trigger

Invoice follow-ups need to be sent at consistent intervals. A 24-hour schedule ensures reminders go out every day without manual effort, keeping payment collection on track automatically.

Dynamic Due Date Calculation

Instead of hardcoding specific reminder days, the Check Due Dates node calculates the difference between today and each invoice due date dynamically. Any invoice with `diffDays >= 1` and `<= 5` is upcoming, `diffDays = 0` is due today, and `diffDays < 0` is overdue — making the system flexible for any due date.

Switch Node for 3-Way Routing

A Switch node routes invoices into three separate email branches based on `reminder_type`. This allows each category to have a different email tone — friendly for upcoming, urgent for due today, and firm for overdue — improving payment response rates.

Log to Reminders Sheet Connected from Switch Node

The Reminders Log node is connected directly from the Switch node (not from Gmail nodes). This ensures original invoice data (`client_name`, email, amount) is always available for logging, since Gmail nodes only output thread IDs and labels — not the original data.

Upsert by Invoice Number

Log to Reminders Sheet uses Update or Insert matched by `invoice_number`. This prevents duplicate log entries when the workflow runs multiple times for testing, ensuring each invoice appears only once in the log regardless of how many runs occur.

Paid Invoice Filtering

Check Due Dates node skips any invoice with `status = "paid"`. This means updating an invoice status to Paid in the sheet is enough to stop all future reminders for that client — no workflow changes needed.

3 Assumptions Made

Invoice Data

- Invoices are manually added to the Google Sheets Invoices tab with correct due dates.
- Due dates are in YYYY-MM-DD format — the JavaScript Date parser handles this correctly.
- Status column is either "Unpaid" or "Paid" — any other value is treated as unpaid.

Email Delivery

- Gmail OAuth2 credentials remain valid for daily automated sending.
- Client emails in the sheet are real and deliverable — no validation is performed.
- Gmail daily sending limits are not exceeded for the volume of invoices.

Business Logic

- Upcoming reminders are sent for invoices due in 1-5 days — not just exactly 3 days.
- All overdue invoices receive reminders daily until marked as Paid.
- No escalation logic is needed — same email template is used for all overdue notices.

4 Error Handling & Edge Cases

Scenario	How It Is Handled
No invoices need reminders today	Check Skip node stops pipeline gracefully with no output
Invoice already paid	Check Due Dates skips any row where status = paid
Missing due date in sheet	Check Due Dates skips rows with empty due_date field
Gmail sending failure	n8n stops on error — operator notified, can retry manually next day
Duplicate log entries from multiple runs	Log to Reminders Sheet uses upsert by invoice_number — no duplicates
Invoice overdue by many days	Dynamic reminder_type (overdue_Ndays) captures any number of days overdue
Log node receiving Gmail data instead of invoice data	Log node connected directly from Switch node — bypasses Gmail output entirely

5 Known Limitations & Future Improvements

Current Limitations

- **No WhatsApp integration:** Only email reminders — WhatsApp Business API requires approval.
- **No escalation logic:** Same email template regardless of how many days overdue.
- **Manual status update:** Client must manually change status to Paid in sheet.
- **No payment portal:** No direct payment link included in reminder emails.
- **Single currency:** Amount field is plain text — no currency conversion logic.

What I Would Improve With More Time

- **WhatsApp reminders:** Multi-channel follow-up via WhatsApp Business API alongside email.
- **Escalation templates:** Different email tone for 1 day, 7 days, and 30 days overdue.
- **Payment link integration:** Include Stripe or PayPal payment link in each reminder.
- **Auto-mark paid:** Webhook from payment processor to automatically update status to Paid.
- **Manager escalation:** Notify business owner when invoice is 30+ days overdue.

6 Technology Stack

Tool	Usage	Reason
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n8n Cloud	Workflow orchestration	Visual builder with code nodes — handles complex routing and scheduling
Google Sheets	Invoice storage and reminders log	Easy to manage invoices manually, full log history visible
Gmail OAuth2	Email reminder dispatch	Reliable delivery, supports personalized message templates
JavaScript (Code node)	Dynamic due date calculation	Calculates exact days difference and assigns reminder type dynamically
Switch node	Three-way email routing	Routes upcoming/due today/overdue invoices to correct email template
Luxon (built-in n8n)	Timestamp formatting	Pakistan timezone timestamps for all reminder log entries

Abdul Rehman Ali — AI Automation Builder & BSCS Student, Air University Kharian
 abdulrehman.tp.786@gmail.com | [linkedin.com/in/abdul-rehman-ali](https://www.linkedin.com/in/abdul-rehman-ali) | abdulrehmanali.netlify.app